

flashjet

GPU-native jet clustering with the full merge history

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Jet clustering, and flashjet

Reconstruction is moving to accelerators — clustering has not

Reconstruction is going heterogeneous

- CMS runs GPUs in the HLT since Run 3
- **~30 %** of HLT reconstruction offloaded
- $\rightarrow$ **~25 %** less total HLT time

pixel local reco · pixel tracks + vertices · calorimeter local reco

Ported so far

- tracking · vertexing · calorimetry
- not jet clustering**

Why it is still on the CPU

- sequential recombination is **IRC safe** — why it survived
- but each merge **depends on the previous one**
- that data dependence does not map onto a GPU in the obvious way

Where it hurts

- reclustering **inside a training loop**
- scanning the **jet radius**
- **substructure** on demand

each one pays a host $\leftrightarrow$ device transfer, over and over

An open-source GPU jet clusterer

- k_t , Cambridge–Aachen, anti- k_t
- custom **Triton** kernels

Why Triton

- the GPU language of the PyTorch ecosystem
- **compiles and autotunes** for the hardware at hand
- **no separate CUDA build**, no device code to maintain

Execution

- one program clusters **one event on chip**
- the batch goes out in **one fused launch**
- jets, histories, substructure → **tensors**

What makes it tractable

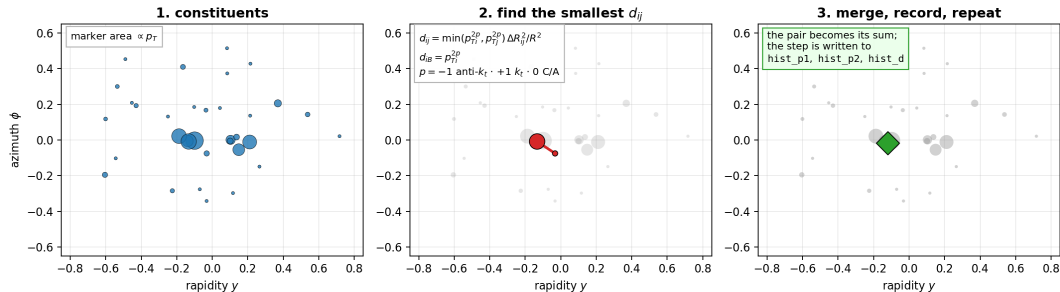
Cacciari–Salam NN lemma: the global d_{ij} minimum sits at some particle's **geometric** nearest neighbour.

- each slot carries its own NN
- a merge invalidates only $O(1)$ rows
- $\Rightarrow O(N^2)$, **not** $O(N^3)$

Engineering

- bandwidth-bound → NN update and **next argmin** fused into one sweep
- **branchless** pair-vs-beam (masked stores)
- autotuned **once per GPU model**, cached → **reproducible**

Sequential recombination — and what flashjet keeps from it



Repeat until nothing is left: if the smallest distance is a d_{ij} , merge that pair; if it is a d_{iB} , call that a jet and remove it. One exponent p selects the algorithm — the *same* kernel serves all three.

One exponent, three algorithms

The generalised- k_t distance measure:

$$d_{ij} = \min\left(p_{Ti}^{2p}, p_{Tj}^{2p}\right) \frac{\Delta R_{ij}^2}{R^2}, \quad \Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$$

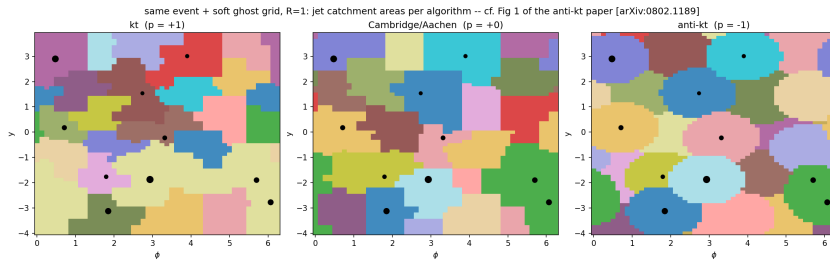
$$d_{iB} = p_{Ti}^{2p}$$

Merge the pair with the smallest d_{ij} ; if the smallest distance is a d_{iB} , call i a jet and remove it.

Only the exponent p changes:

p	algorithm	merges first	the tree is ordered by
-1	anti- k_t	around the hardest particles	— (gives the jets)
0	Cambridge–Aachen	the closest pair	angle
+1	k_t	the softest pair	relative p_T

The ordering is **what the substructure reads**: C/A is the tree for **grooming** and the **Lund plane**, k_t for **exclusive subjets**.



Jet catchment areas for the three algorithms, from flashjet's own clustering. k_t and C/A give ragged, fluctuating boundaries; **anti- k_t gives circles of area πR^2** around each hard particle — the defining result of the algorithm, recovered here without reference to FastJet.

It returns the tree, not just the jets

Clustering a batch returns **two** things — the jets, and the record of how they were built.

field	shape	what it holds
jet_idx	(B, N)	which jet each particle ended up in
n_jets	$(B,)$	how many jets were found
hist_p1, hist_p2	(B, N)	which pair merged , at each step
hist_child	(B, N)	what they merged <i>into</i>
hist_d	(B, N)	the distance d_{ij} at that step

Why the second half matters

The bottom three rows are the **complete binary merge tree**. Groomed mass, z_g , R_g , Lund coordinates and exclusive subjets are all **coordinates on that tree** — so they become array reads, with **no second clustering pass**.

Using it

```
import flashjet

out = flashjet.cluster(p4, mask, R=1.0, algorithm="antikt")
```

argument	type	
p4	$(B, N, 4)$ tensor	(p_x, p_y, p_z, E) ; CPU or CUDA
mask	(B, N) bool	marks real constituents in a padded batch
R	float	jet radius
algorithm	str	"antikt" "kt" "cambridge"
p	float	generalised- k_t exponent (overrides algorithm)

out.n_jets	# (B,)	jets found
out.jets_p4(p4)	# (B,J,4)	jet four-momenta
out.splitting_scales()	#	$\sqrt{d_{12}}$, $\sqrt{d_{23}}$, ...
out.exclusive_jets(n_jets=3)	# F1	exclusive k_t subjects
out.groomed_jets(p4, R, z_cut=0.1, beta=0.0)	# F2	soft drop / mMDT
out.lund_coordinates(p4, R)	# F3	primary Lund plane

The same call runs unchanged on CPU and GPU — the backend is chosen from the problem size, not by the caller.

What is implemented

Clustering

- generalised- k_t : anti- k_t , k_t , C/A
- batched, ragged input via a mask
- three backends, auto-dispatched by size
- merge history recorded in-kernel
- jet regime *and* event regime

Substructure — reads of the history

- **F1** exclusive k_t subjets
- **F2** soft drop / mMDT / mass-drop
- **F3** primary Lund coordinates

All three are pure-torch post-reads: no kernel changes, CPU/GPU identical, negligible cost.

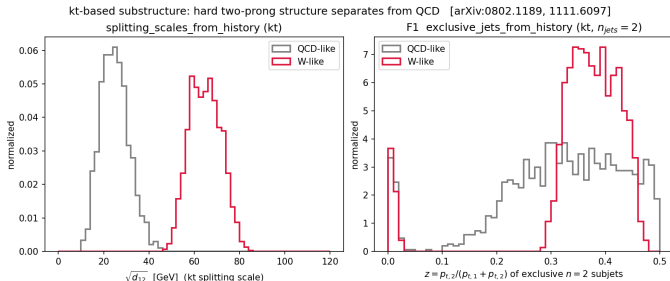
What it does. Undo the last merges of the sequence to expose exactly n subjects — or stop at a d_{cut} .

How. The history is already a binary tree. Walk back up it $n - 1$ steps; no reclustering.

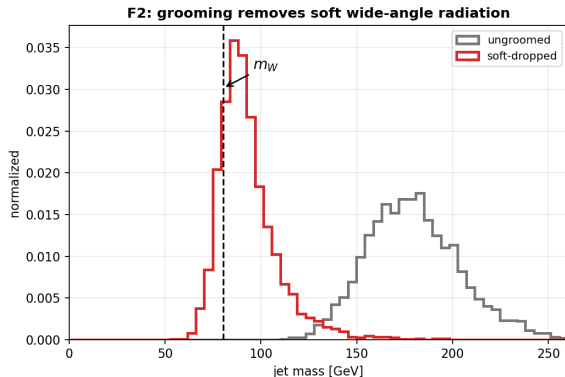
Why k_t . k_t merges the softest pair first, so the *last* merges are the hard prongs — exactly what a 2-prong decay leaves behind.

The splitting scales come straight from `hist_d`:

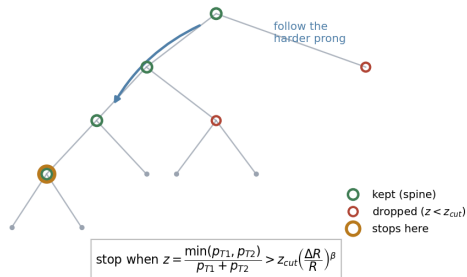
$$\sqrt{d_{12}}, \sqrt{d_{23}}, \dots \text{ with } d_{ij} = \min(p_{Ti}^2, p_{Tj}^2) \Delta R_{ij}^2 / R^2$$



Toy W-like (2-prong) vs QCD-like fat jets. **Left:** the k_t splitting scale $\sqrt{d_{12}}$ separates cleanly. **Right:** the momentum balance z of the two exclusive subjects — the W sits near $z \sim 0.4$, QCD spreads to small z .



the soft-drop walk on the C/A tree



Walk down the C/A tree along the harder branch, discarding the softer prong until the momentum-sharing condition is met. $\beta = 0$ recovers mMDT.

The walk is $O(\log_2 n)$ on the stored tree — it never touches the constituents again, so grooming is essentially free once clustered.

C/A merge trees of real jets — root at top, constituents at the bottom

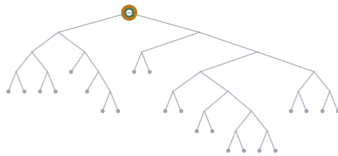
ATLAS Top Tagging Open Data · vertical = declustering depth · horizontal = angular ordering

Light quark / gluon jet



a long spine: soft prong after soft prong
is stripped, and the mass collapses
 $n_{\text{drop}} = 8$ $m: 80 \rightarrow 1 \text{ GeV}$

Boosted top — a clean two-prong core



the very first split is already balanced,
so nothing is groomed away
 $n_{\text{drop}} = 0$ $m: 80 \rightarrow 80 \text{ GeV}$

Boosted top — the full $t \rightarrow bW$ system



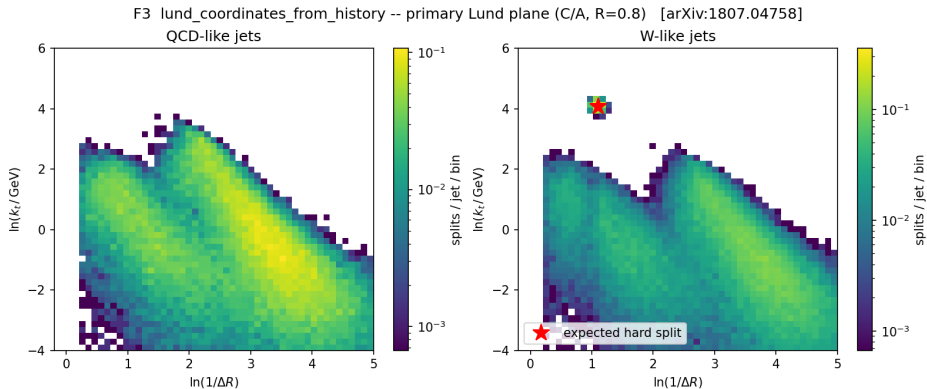
also stops at once, but on a wider split,
so the whole decay is kept
 $n_{\text{drop}} = 0$ $m: 170 \rightarrow 170 \text{ GeV}$

● soft-drop spine ● dropped prong ● grooming stops here ● constituent

A jet without a decay has nothing to stop on. The light-quark/gluon jet is a long *staircase*: soft prong after soft prong is stripped, and its mass collapses to nothing.

ATLAS Open Data, C/A with soft drop ($z_{\text{cut}}=0.1$, $\beta=0$). **The number of drops is itself a discriminant** — and it is read straight off the stored tree, at no extra cost.

A real decay stops at once. Both top jets halt on their very first split, because it is already hard and balanced — so the mass survives.



What. Every primary splitting emits $z = \frac{p_{T2}}{p_{T1}+p_{T2}}$, ΔR , $k_t = p_{T2}\Delta R$, and the plane coordinates $(\ln \frac{1}{\Delta R}, \ln k_t)$.

QCD fills the soft-collinear region smoothly; the W-like sample adds a **localised hard splitting** exactly where the decay is predicted to sit (star).

How. One pass down the primary branch — each node is one point in the plane.

Check. The d -channel ties *exactly* to the splitting scales computed independently.

Correctness

Two kinds of validation

First: toys, where truth is known

On generated inputs the *answer* is known analytically — so any observable landing off the predicted curve is a real bug, with nothing to hide behind.

Input A — toy fat jets — QCD-like and W-like fat jets

Input B — toy parton shower — leading-log parton shower

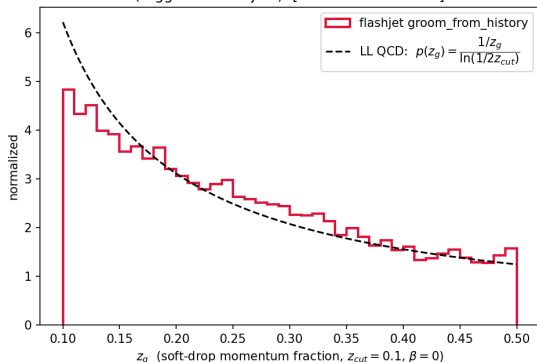
Then: real data, against FastJet

On detector-level jets there is no analytic answer — so the reference is **FastJet on the same events**, and the experiment's own reconstructed jets.

Input C — ATLAS Open Data — large- R jets and full events

The order matters: **toys first** pins the algorithm against a known answer; **real data second** shows it survives detector-level input. A failure at either stage means something different.

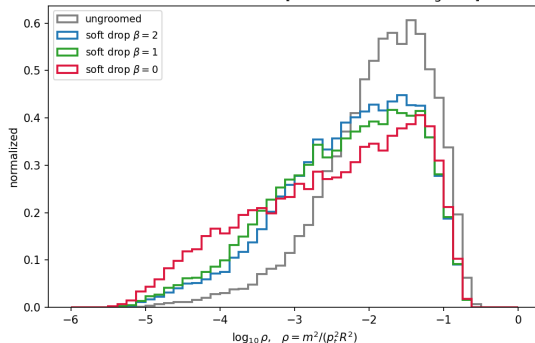
groomed splitting fraction vs the analytic prediction
(tagged 72% of jets) [cf. arXiv:1402.2657]



Soft-drop z_g follows the $1/z$ form predicted at leading-log, recovered from the C/A history.

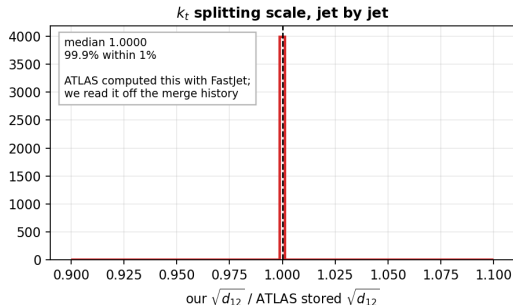
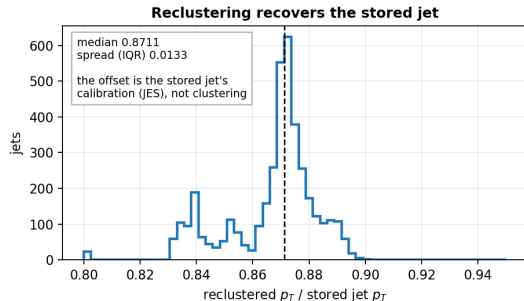
Toy-shower input, so the *prediction* is unambiguous — these test the decoders, not an event generator.

groomed jet mass: stronger grooming (smaller β) pushes the soft-radiation mass down [cf. arXiv:1402.2657 Figs 3-4]



Groomed-mass ordering in β : larger β grooms less, so the mass peak moves up.

Jet-by-jet closure against ATLAS-stored quantities



Left. Reclustering the stored constituents recovers the stored jet with a **0.013 spread**. The 0.87 offset is the stored jet's **calibration**, applied after clustering — it is present before we touch anything.

Right. The k_t splitting scale $\sqrt{d_{12}}$ against the value **ATLAS computed with FastJet** and stored: median **1.0000**, **99.9 %** within 1%. We read ours straight off the merge history.

3 algorithms $\times$ **5 radii** $\times$ **5 multiplicity bins** $\times$ **2 samples**, 20 000 jets per bin, on ATLAS Open Data.

check	result
number of jets found	100.000 % at every one of 150 points
leading-jet p_T within 10^{-4}	1.0000 at 148/150 points
median relative difference	$\sim 7 \times 10^{-8}$ (float32 precision)

The grid is the point

A single working point proves little. Agreement holds across **all three algorithms**, **every radius** from 0.4 to 1.2, and **24 to 123** constituents per jet.

Two independent references

The same events are clustered by **FastJet** for the jets, and every substructure decoder is separately pinned against an **independent NumPy tree-walk**.

A stronger test than agreeing with FastJet.

ATLAS publishes an open dataset carrying both the **calorimeter clusters** and **its own reconstructed jets**. Cluster the ~ 600 clusters per event ourselves, and compare:

n_{jets} vs ATLAS	100.0 % match
mean jets/event	10.96 vs 10.96

This closes the algorithm against **the experiment's own published output** — not against another implementation of the same algorithm.

599.3 clusters/event: the event regime, where the problem is $O(N^2)$.

Speed

Data — ATLAS Open Data, freely presentable

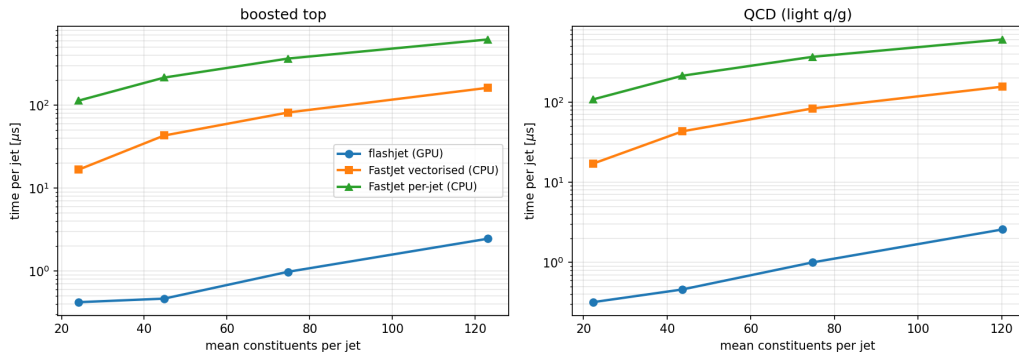
sample	jets	$\langle n_{\text{const}} \rangle$
boosted top	30 000	59.2
QCD (light q/g)	30 000	52.4

Large- R jets with calorimeter-cluster constituents. The grid is **3 algorithms** $\times$ **5 radii** $\times$ **5 multiplicity bins**, 20 000 jets per bin.

Method

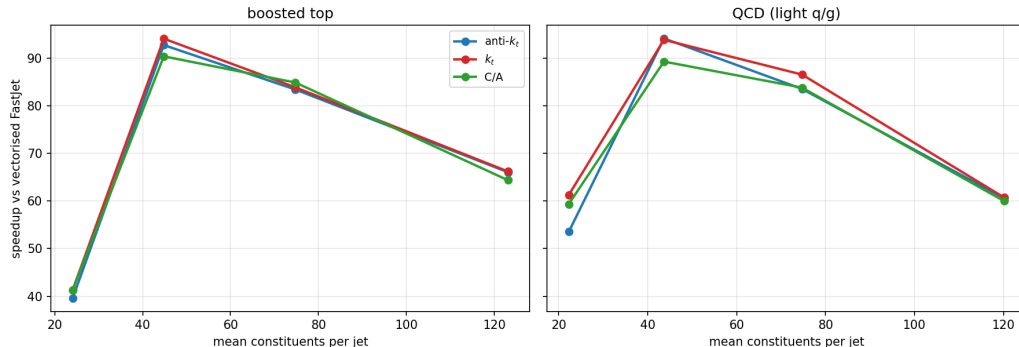
- 1 extract constituents once, save;
- 2 cluster on GPU, time it, save the *same* events;
- 3 cluster those saved events with FastJet;
- 4 **check they agree** — else the timing is meaningless.

Both clusterers read the **same saved file**. Speedups are quoted against the **vectorised** FastJet interface — the faster, fairer baseline.

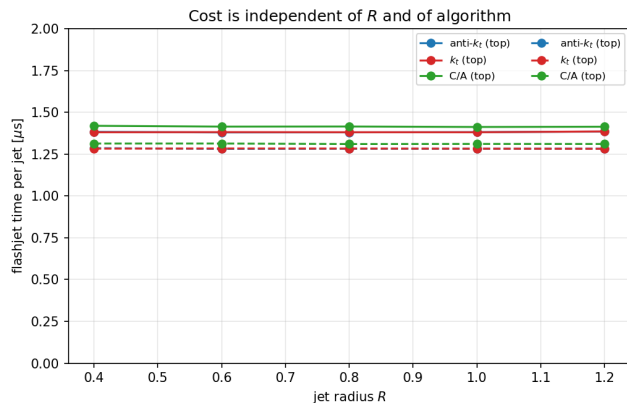
ATLAS Top Tagging Open Data — anti- k_t $R = 1.0$ 

Log scale — each gridline is $10\times$. flashjet stays nearly flat while both FastJet interfaces climb steeply with multiplicity: **0.35–2.7 μ s per jet** against **17–155 μ s** for vectorised FastJet, and **110–620 μ s per jet** one at a time. Benchmarked on an NVIDIA V100-class GPU.

Speedup vs multiplicity — $R = 1.0$ (all points 100% n_{jets} agreement)



All three algorithms, both samples, $R = 1.0$. **39–99 $\times$** over vectorised FastJet across the grid (median **64 $\times$**), and every point has **100 %** n_{jets} agreement. Busier jets $\rightarrow$ bigger win — the regime that matters for boosted-object physics.



Across $R = 0.4 \rightarrow 1.2$: a **0.3 % spread** over a $3\times$ range in R .

R changes *which* pairs merge, not *how many* distances are computed.

Across algorithms: anti- k_t , k_t and C/A agree to within $\sim 4\%$ at every multiplicity.

k_t and C/A cost the same as anti- k_t .

Downstream: the clustering tree as tagger input

[to be filled in]

ParT training setup

[to be filled in]

ParT training result

- 1 **Clusters on the GPU and keeps the full merge history**, so substructure — exclusive subjects, soft drop, Lund coordinates — comes free as array reads.
- 2 **Gives the same jets as FastJet.** 150/150 grid points at 100 % n_{jets} agreement, across anti- k_t/k_t /C-A, $R = 0.4\text{--}1.2$ and 24–123 constituents per jet.
- 3 **Reproduces the experiment's own reconstructed jets** exactly, at ~ 600 clusters per event.
- 4 **39–99 $\times$ faster** than vectorised FastJet, and the advantage *grows* with jet complexity. Cost is independent of R and of algorithm.
- 5 Still a **prototype**, with meaningful headroom left.

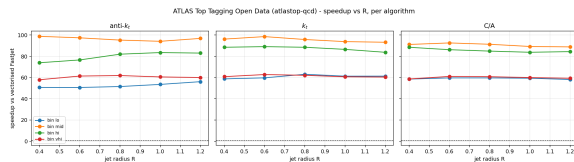
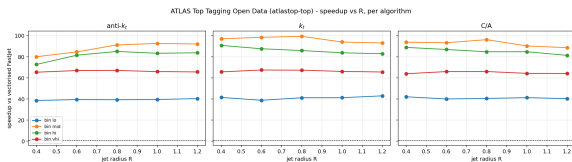
Thank you

Questions?

Benchmarks: ATLAS Open Data (CC0).

Physics closures: analytic predictions on toy showers.

Backup

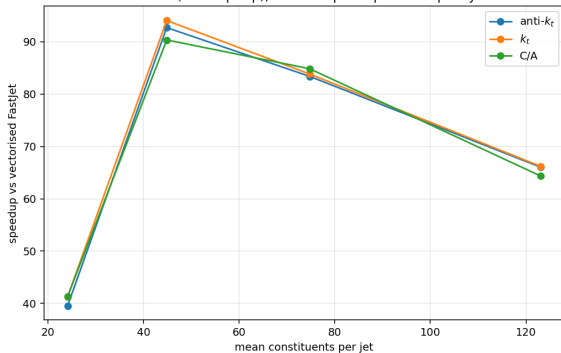


boosted top

Flat in R for all three algorithms; ordering is by multiplicity bin.

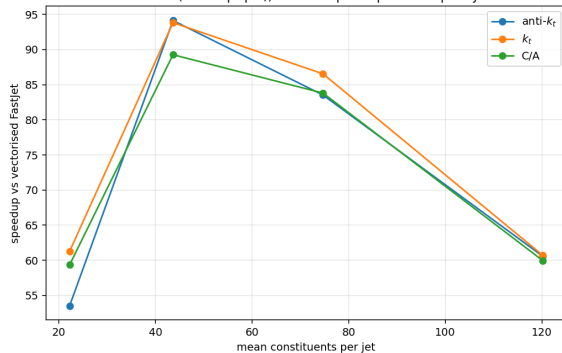
QCD (light q/g)

ATLAS (atlastop-top), R=1.0 - speedup vs multiplicity



boosted top

ATLAS (atlastop-qcd), R=1.0 - speedup vs multiplicity



QCD (light q/g)

Backup — the numbers

sample	alg	$\langle n \rangle$	flashjet [$\mu\text{s}/\text{jet}$]	FastJet vec. [$\mu\text{s}/\text{jet}$]	speedup
top	anti- k_t	24.1	0.427	16.5	39×
top	anti- k_t	44.8	0.464	43.0	93×
top	C/A	74.7	1.000	84.8	85×
top	C/A	123.1	2.565	165.1	64×
qcd	anti- k_t	22.2	0.346	17.5	51×
qcd	anti- k_t	43.6	0.463	45.7	99×
qcd	anti- k_t	74.7	1.215	89.8	74×
qcd	anti- k_t	120.2	2.673	154.9	58×

39–99× over vectorised FastJet (median 64×), with 100 % jet-count agreement everywhere. Benchmarked on an **NVIDIA V100-class** GPU.

$R = 1.0$ rows shown. Against the per-jet FastJet interface the gap is roughly *two orders of magnitude*.

Backup — where the time goes

Nsight Systems — share of GPU time

kernel	% GPU	avg/call
_cluster_large_kernel	97.6	5.47 ms
_decode_kernel	1.5	82 μ s
torch glue	< 1	—

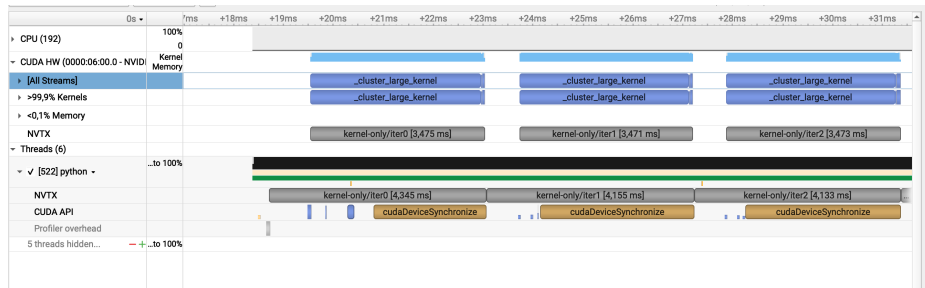
No hidden overhead — essentially all the time is real clustering work, and the memory row of the timeline is empty.

Nsight Compute — hardware counters

metric	value
DRAM throughput	0.02 %
Compute (SM) throughput	15.4 %
Registers / thread	168
Theoretical occupancy	18.75 %
Achieved occupancy	6.25 %
Waves per SM	0.32

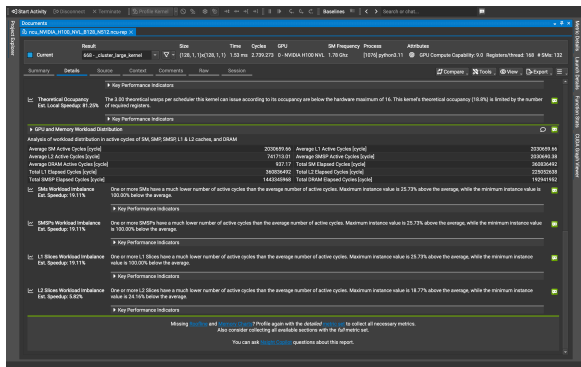
Not memory- or compute-bound — the GPU is simply **not filled**.

Backup — the GPU timeline



Nsight Systems, three consecutive iterations. Each block is one clustering call; the tool labels the rows “>99.9 % Kernels” and “<0.1 % Memory”. The kernels run back to back with nothing in between and the memory row is empty — no host/device shuffling while clustering runs. That is what makes the speedup real rather than an artefact of where the timer was started.

Backup — the profiler names the bottleneck



The tool's own speedup estimates:

finding est. speedup

Theoretical occupancy (registers) 81%

Achieved occupancy 67%

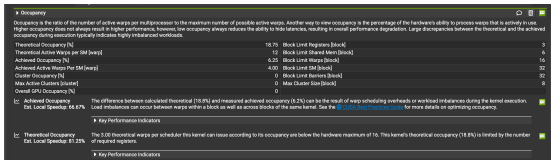
SM workload imbalance 19%

"This kernel grid is too small to fill the available resources on this device, resulting in only 0.32 full waves across all SMs."

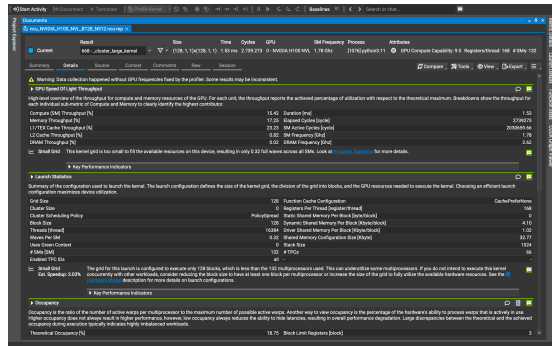
Grid size **128 blocks** on a **132-SM** device — fewer blocks than the GPU has processors.

~**1.7–1.8×** more available from launch configuration and register budget — **not** from the algorithm.

Backup — occupancy detail



Occupancy		
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.		
Theoretical Occupancy (%)	18.75	Block Limit Registers [Block]
Theoretical Active Warps per SM [warp]	15	Block Limit Shared Mem [Block]
Achieved Occupancy (%)	6.25	Block Limit Warps [Block]
Achieved Active Warps Per SM [warp]	4.00	Block Limit SM [Block]
Cluster Occupancy (%)	0	Block Limit Registers [Block]
Max Active Clusters [cluster]	0	Max Cluster Size [Block]
Overall GPU Occupancy (%)	0	
The difference between calculated theoretical (18.75%) and measured achieved occupancy (6.25%) can be the result of warp scheduling overheads or workload imbalances during the kernel execution. Load imbalances can occur between warps within a block as well as across blocks of the same kernel. See the Occupancy Performance Guide for more details on optimizing occupancy.		
Key Performance Indicators		
The 3.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 15. This kernel's theoretical occupancy (18.8%) is limited by the number of required registers.		
Key Performance Indicators		

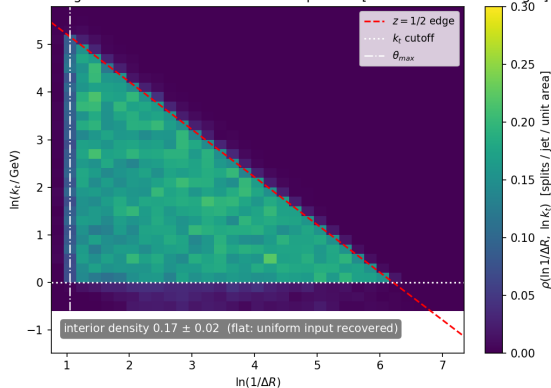


Result		Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
808	1-cholster_large_kernel	1108, 1, 1 (x1028, 1, 1)	1.83 ms	2,739,273	0 - M1050A.H100 M1, 1.18 GHz	117670	python3.11	GPU compute Capability 8.8 Registers/Thread: 168 # Sm: 102

Summary		Details	Source	Context	Comments	Raw	Session
Warning: Data collection happened without GPU frequencies being set by the profiler. Some results may be inaccurate.							
GPU Speed (Of Light) Throughput							
High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-items of Compute and Memory to clearly identify the highest constraint.							
Compute (SM) Throughput (%)	15.40	Duration [ms]	1.53				
Memory Throughput (%)	17.25	Elapsed Cycles [cycles]	2739273				
L1/L2 Cache Throughput (%)	23.23	SM Active Cycles [cycles]	203048566				
L2 Cache Throughput (%)	0.82	SM Frequency [GHz]	1.18				
DRAM Throughput (%)	0.02	DRAM Frequency [GHz]	2.62				
Small Grid - This kernel grid is too small to fill the available resources on this device, resulting in only 6.32 full warps across all SMs. Look at Launch Statistics for more details.							
Key Performance Indicators							
Launch Statistics							
Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.							
Grid Size	128	Function Cache Configuration	CachePrefetchMode				
Cluster Size	0	Registers Per Thread [register/thread]	168				
Cluster Scheduling Policy	Polygraph	Static Shared Memory Per Block [Byte/block]	0				
Block Size	128	Dynamic Shared Memory Per Block [Byte/block]	4.10				
Threads [thread]	16384	Driver Shared Memory Per Block [Byte/block]	1.02				
Waves Per SM	0.32	Shared Memory Configuration Size [Byte]	32.77				
Waves Per Cluster	0	Bank Size	16384				
# SMs (SM)	102	# TPCs	66				
Enabled TPCs	all						
Small Grid - The grid for this launch is configured to execute only 128 blocks, which is less than the 102 multiprocessors used. This can underutilize some multiprocessors. If you do not intend to execute this kernel concurrently with other workloads, consider reducing the block size to have at least one block per multiprocessor or increase the size of the grid to fully utilize the available hardware resources. See the Workload Model description for more details on launch configurations.							
Key Performance Indicators							
Occupancy							
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.							
Theoretical Occupancy (%)	18.75	Block Limit Registers [Block]					

Register pressure (168/thread) caps theoretical occupancy at 18.75 %; achieved is 6.25 %. With 0.32 waves/SM the kernel does not fill the GPU even once, while DRAM throughput is 0.02 % — this is parallelism starvation, not a bandwidth problem. Both are configuration inherited from smaller GPUs, not properties of the algorithm.

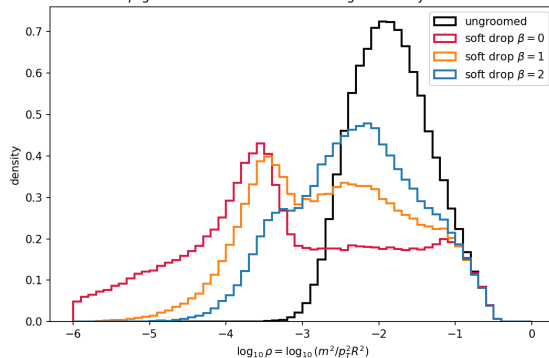
primary Lund plane of the toy shower (C/A, R=0.4):
m-in-triangle emissions come back as a flat plateau [cf. arXiv:1807.04758 Fig 2]



The Lund plane is populated **uniformly** inside the kinematic **triangle** — the leading-log expectation.

Independent NumPy tree-walks pin every decoder, and the d -channel of the Lund output ties *exactly* to the splitting scales computed separately.

Soft-drop β -ordering on REAL QCD jets (164292 jets, $z_{cut} = 0.1$)
smaller β grooms harder — same ordering as the toy-shower closure



The soft-drop β -family orders as predicted: larger β grooms less.

Backup — Input A: the toy fat jets

Purpose. QCD vs W is a *known* truth, so any observable that fails to separate them — or lands off the analytic curve — is a real bug.

Two single-fat-jet samples at $p_T \approx 500$ GeV, $R = 0.8$:

- **QCD-like:** one hard collinear core (92 % of p_T , $\sigma_{y,\phi} = 0.06$) + soft wide-angle radiation (8 %, $\sigma = 0.30$)
- **W-like:** two hard prongs with pair mass $m = \sqrt{z(1-z)} p_T$, $\Delta R = 80.4$ GeV, $z \in [0.30, 0.45]$, random orientation, + 6 % soft contamination

Each “spray” is n massive pions with exponential p_T fractions:

```
frac = rng.exponential(1., n); frac /= frac.sum()
pt    = pt_total * frac
y, phi = rng.normal(y0, sigma, n), rng.normal(phi0, sigma, n)
mt     = np.sqrt(M_PI**2 + pt**2)
```

1500 events per sample. flashjet has **no event generator** — it clusters four-vectors; these toys live in the plot scripts, outside the library.

Backup — Input B: the toy parton shower

A **primary, fixed-coupling, leading-log** shower: emissions sampled **uniformly in the Lund triangle** with density $\bar{\alpha}$ per unit $(\ln 1/\theta, \ln k_t)$ area, off a hard spine.

$$\theta = e^{-u}, \quad k_t = e^v, \quad z = \frac{k_t}{p_{T0} \theta}$$

keep $v \leq \ln(p_{T0}/2) - u$ (i.e. $z \leq \frac{1}{2}$), $k_t > k_t^{\min}$

Why this is the right test

The generator's density is *uniform in the triangle by construction*, so the leading-log predictions — the $1/z$ form of z_g , the β -ordering of the groomed mass — are unambiguous. Landing on them tests the **decoders**, not the shower.

Jet areas use the anti- k_t -paper construction: one hard event plus a dense grid of **infinitely soft ghosts** ($p_T = 10^{-8}$); the ghosts each algorithm sweeps up trace its catchment area.

$\bar{\alpha} = 0.25$, $p_{T0} = 1$ TeV, $R = 0.4$, 20 000 showers; fixed seed, fully re-runnable.

Backup — the validation ladder

rung	input	reference	result
unit tests	—	independent NumPy tree-walks	129 pass, incl. GPU paths
anti- k_t jet areas	toy B	the anti- k_t paper's own construction	rigid cones, $\rightarrow \pi R^2$
k_t substructure	toy A	known 2-prong vs QCD truth	clean separation
soft-drop z_g	toy B	leading-log $1/z$ prediction	on the curve
groomed mass vs β	toy B	Soft Drop Figs. 3–4	ordering reproduced
Lund plane	toy A	primary-Lund construction	hard split where predicted
jet-by-jet kinematics	real C	FastJet, same events	150/150 points, 100 % n_{jets}
full-event clustering	real C	the experiment's own jets	100 % match, ~ 600 clusters/ev

Each rung is closed against something independent — an analytic prediction, a separate NumPy implementation, or FastJet itself. No rung is validated against another part of flashjet.

All benchmark data is **ATLAS Open Data** (CC0), read over the public redirector — no grid certificate required.

dataset	used for
ATLAS Top Tagging Open Data	timing + FastJet agreement
2020 ATLAS Jet Reconstruction	closure vs the experiment's own jets

Constituents are calorimeter clusters, massless. Physics closures use toy showers compared against leading-log analytic predictions, so the prediction being tested is unambiguous.